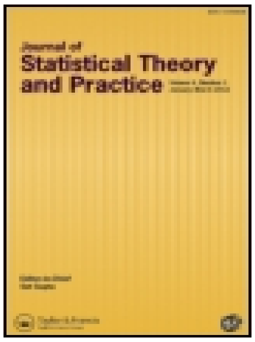

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Abstract

In this paper, we propose a new integer-valued autoregressive process with generalized Poisson difference marginal distributions based on difference of two quasi-binomial thinning operators. This model is suitable for data sets on $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots$ and can be viewed as a generalisation of the Poisson difference INAR(1) process. An advantage of the difference of two generalized Poisson random variables is it can have longer or shorter tails compared to the Poisson difference distribution. We present some basic properties of the process like mean, variance, skewness and kurtosis, conditional properties of the process are derived. Yule Walker estimators are considered for the unknown parameters of the model and a Monte Carlo simulation is presented to study the performance of estimators. An application to a real data set is discussed to show the potential for practice of our model.

Keywords: Generalized Poisson process; quasi-binomial distribution; Integer-valued time series.

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1. Introduction

In the last three decades several authors have discussed models for integer-valued time series. Al-Osh and Alzaid (1987) introduced the first-order integer-valued autoregressive (INAR(1)) process with Poisson marginal using the binomial thinning operator introduced by Steutal and Van Harn (1979). Alzaid and Al-Osh (1990) generalized the INAR process, defining an integer-valued p th-order autoregressive structure, INAR(p) process, in the same way of INAR(1) process. Du and Li (1991) proposed the integer-valued autoregressive INAR(p) model using the binomial thinning operator and added the hypotheses that the innovations series independent of the counting series. A first-order autoregressive process with generalized Poisson marginal distribution (GPAR(1)) was defined by Alzaid and Al-Osh (1993) based on the quasi-binomial thinning operator. Ristić et al. (2009) introduced a count valued time series model with geometric marginal with negative binomial thinning operator. This model is an alternative to the Poisson INAR(1) when the data set is overdispersed. Several other models were defined for time series with non-negative values with different thinning operators (see Weiß, 2008). In many fields it is common to have data sets based on negative and positive values, such as medicine (Karlis and Ntzoufras, 2006), financial studies(Shahtahmassebi and Moyeed, 2013, Chesneau and Kachour, 2012), sports (Shahtahmassebi and Moyeed, 2016). Thus, it becomes necessary to define integer-valued time series models for such specific data sets. Recently, many authors have presented time series models when the positive and negative integer values arise as the difference between two discrete distributions. Freeland (2010) defined a first-order stationary autoregressive process on $\mathbb{Z}$ with symmetric Skellam marginals, the new process was defined by a difference between two independent and identically distributed random variables with Poisson distribution and a new thinning operator obtained by the difference of two binomial thinning operators, the process was called True INAR(1) (TINAR(1)). Barreto-Souza and Bourguignon (2015) defined a modified negative binomial thinning operator by the difference of two negative binomial thinning operators and defined an INAR(1) process on $\mathbb{Z}$ with skew discrete Laplace marginals by the difference of two independent NGINAR(1) (Ristić et al., 2009) process with geometric marginals with means $\mu_1 > 0$ and $\mu_2 > 0$, respectively. Some properties of the process were derived, including joint and conditional basic properties, characteristic function, moments and estimators for the parameters. Another thinning operator using the difference of two negative binomial thinning operators was defined by NÃ¡stic et al. (2016). Based on the difference of two independent random variables with geometric marginal distributions with the same mean $\mu > 0$, they defined a new symmetric INAR(1) process with discrete Laplace marginal distributions with positive or negative lag-one autocorrelation. The properties of the new thinning operator were derived and properties of the process were obtained. Alzaid and Omair (2014) defined a new operator called the extended binomial thinning operator and introduced the Poisson difference integer valued autoregressive model of order one, PDINAR(1) process, with Poisson difference marginal, Skellam distribution, to model integer valued time series with possible negative values and either positive or negative correlations. This model was applied to data from the Saudi stock exchange. A stationary first-order integer valued autoregressive process with geometric-Poisson marginals (NSINAR(1)) was introduced by Bourguingnon and Vasconcellos (2016). The new process allows negative values in the series. Several properties were established and estimators are defined for unknown parameters by the Yule-Walker method. This process was defined as the difference between geometric and Poisson random variables and the new thinning operator based on the difference of the negative binomial thinning operator and the binomial thinning operator.

In this paper, we propose a new thinning operator based on the idea of Freeland

(2010), using the sequence of quasi-binomial thinning operator defined by Alzaid and Al-Osh (1993). Then we define the skew integer valued autoregressive process with generalized Poisson difference marginal distribution (GPDAR(1)) as the difference between two random variables that follow generalized Poisson (GP) distribution. The motivation for such a process arises from the necessity to have adequate models that can be adapted to skew integer valued (including both negative and positive values) time series. Moreover, this process with generalized Poisson difference (GPD) marginal has more flexibility in the tails than the process with Poisson difference (PD) marginal distribution, that, in some cases, has a tendency to underestimate values in the tails (see Shahtahmassebi and Moyeed, 2014). The distribution of the difference of two GP random variables can have longer or shorter tails compared to the PD distribution and the GPD distribution can be viewed as a generalisation of the PD distribution.

The paper is organized as follows: In Section 2, we present a review of the GP and quasi-binomial (QB) distributions, and the GPAR(1) process. The GPDAR(1) process is defined based on the difference of two quasi-binomial thinning operators. In Section 3, some of its properties are derived. In Section 4, the Yule-Walker estimators of unknown parameters are presented and numerical results using Monte Carlo simulation are discussed. In Section 5, an application to real life data set is presented. Finally, we conclude the paper in Section 6.

2. GPDAR(1) process

In this section, we present a stationary first-order integer-valued autoregressive process with generalized Poisson difference marginals. To that end, we first present brief reviews of the GP distribution (Consul and Jain, 1973), QB distribution (Consul and Mital, 1975) and the GPAR(1) process.

2.1. Generalized Poisson distribution

Consul and Jain (1973) defined a new distribution as a generalization of the Poisson distribution, the GP distribution with two parameters. It was obtained as a limiting form of the generalized negative binomial distribution. Consul (1989) discussed the properties and applications of the GP distribution in the book *Generalized Poisson Distribution: Properties and Applications*. For example, in that book, the author described the application of GP distribution to the number of chromosome aberrations induced by chemical and physical agents in human and animal cells. Johnson, Kotz and Kemp (1992) have also described some properties of the GP distribution in the book *Univariate Discrete Distributions*. Joe and Zhu (2005) proved that the GP distribution is a mixture of Poisson distributions and they made a study comparison between the GP and negative binomial distributions. A detailed presentation and discussion of the GP distribution can be found in the book *Lagrangian Probability Distributions* by Consul and Famoye (2006).

The probability mass function (pmf) of a random variable (r.v.) X following a generalized Poisson distribution with parameters λ and θ, denoted by GP(λ,θ), is given by

$$P(X=x)=\lambda(\lambda+\theta x)^{x-1}e^{-(\lambda+\theta x)}/x!, \quad x=0,1,2,\ldots, \tag{1}$$

where $\lambda>0$ and $\theta<1$. For more details, see Consul and Famoye (2006).

Remark 1. Note that for $\theta = 0$, we obtain the Poisson distribution with parameter λ as a special case of the generalized Poisson distribution.

We can obtain some properties about the GP distribution from Johnson, Kotz and

Kemp (2005). We have that the probability generating function (pgf) is given by

$$G(s) = e^{\lambda(t-1)}, \quad \text{where} \quad t = se^{\theta(t-1)},$$

and the mean, variance, third and fourth central moments are given by

$$\mathrm{E}(X) = \frac{\lambda}{(1-\theta)}, \quad \mathrm{Var}(X) = \frac{\lambda}{(1-\theta)^3}, \tag{2}$$

$$\mu_3 = \frac{\lambda(1+2\theta)}{(1-\theta)^5}, \quad \mu_4 = \frac{3\lambda^2}{(1-\theta)^6} + \frac{\lambda(1+8\theta+6\theta^2)}{(1-\theta)^7}.$$

The coefficients of skewness and kurtosis are, respectively,

$$\beta_1 = \frac{(1+2\theta)^2}{\lambda(1-\theta)}, \quad \beta_2 = 3 + \frac{1+8\theta+6\theta^2}{\lambda(1-\theta)}. \tag{3}$$

Figure 1 displays the skewness and kurtosis of GP distribution for fixed $\theta = -0.5, -0.2, 0.1, 0.4, 0.8$ and $\lambda \in \{1,3,5,7,9,11,13,15\}$.

Note that the skewness of the GP distribution depends on the values of λ for any fixed value of θ. As the value of λ increases, the skewness of GP distribution decreases and for λ enough large the skewness converge to 0. For any fixed value of λ, the skewness becomes large for θ values close to 1. Also, observe that both curves have the same shape. This is expected, since $(\beta_2 - 3)/\beta_1$ does not depend on λ.

Remark 2. From (2) the mean of the GP distribution is smaller than the variance when 0 < θ < 1 , which means the GP distribution, in this case, may be used as a model to overdispersed data set. On the other hand, the mean is larger than the variance for $\theta < 0$, which shows that the GP distribution may be also used to model underdispersion. Furthermore, from (3), the GP distribution is adequate to model a skewed data set.

Consider two independent random variables X and Y with $\mathrm{GP}(\lambda_1, \theta_1)$ and $\mathrm{GP}(\lambda_2, \theta_2)$ distributions, respectively, such as (1). The distribution of the difference Z between the two variables is a distribution with support in the set $\mathbb{Z}$ of all integer numbers with probabilities given by

$$P(Z=z) = e^{-\lambda_1 - \lambda_2 - z\theta_1} \sum_{y=y_0}^{\infty} (\lambda_2, \theta_2)_y (\lambda_1, \theta_1)_{y+z} e^{-y(\theta_1+\theta_2)}, \quad z \in \mathbb{Z}.$$

where $y_0 = \max\{0, -z\} = -\min\{0, z\}$ and $(\lambda, \theta)_y = \lambda(\lambda + \theta y)^{y-1} / y!$. We consider the particular case of $\theta_1 = \theta_2 = \theta$, where $0 \le \theta < 1$, which has been studied before by Consul (1986). For two independent random variables X and Y with $\mathrm{GP}(\lambda_1, \theta)$ and $\mathrm{GP}(\lambda_2, \theta)$ distributions, respectively, with pmf such as (1), with $0 \le \theta < 1$, Consul considered the distribution of the difference $Z = X - Y$ and obtained its properties as pmf, cumulants and pgf. We will refer to the distribution proposed by Consul (1986) as the "generalized Poisson difference distribution (GPD)". The generalized Poisson difference distribution with parameters $\lambda_1, \lambda_2, \theta$ ($\mathrm{GPD}(\lambda_1, \lambda_2, \theta)$) has pmf given by

$$P(Z=z) = e^{-\lambda_1 - \lambda_2 - z\theta} \sum_{y=0}^{\infty} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z} e^{-2y\theta}, \quad z \in \mathbb{Z},$$

where $y_0 = \max\{0, -z\} = -\min\{0, z\}$. Properties of this distribution will be presented in

Section 3

In Figure 2, we can observe that for $\lambda_2 > \lambda_1$ the GPD tends to be left skewed andcolorred, as the value of λ_2 increases, the GPD tends to have a longer tail. Also note that the following type of symmetry holds for the GPD distribution:

$$P(Z = z; \lambda_1, \lambda_2, \theta) = P(Z = -z; \lambda_2, \lambda_1, \theta).$$

Than, the GPD behaviour for $\lambda_1 > \lambda_2$ is symmetric to for $\lambda_1 < \lambda_2$.

In the case of $\lambda_1 = \lambda_2$ we have a symmetric process with zero mean as will see in Section 3. Note, from Figure 3, that the variance of the distribution increases as we increase the common value of λ_1 and λ_2.

From Figure 4 we can see that when θ increases the tails of the GPD distribution become heavier.

Finally, Figure 5 shows, for the particular case of $\theta = 0$, shows the behaviour of skewness, depending on the relation between λ_1 and λ_2.

2.2. Quasi-binomial distribution

The QB distribution was introduced by Consul (1974) as an urn model and he showed that the model is a true probability. A new urn model and a general expression to QB distribution was presented by Consul and Mital (1975). Consul (1975) considered the characterization of Lagrangian Poisson and quasi-binomial distributions and the relation between them. Fazal (1976) developed an optimal asymptotic hypothesis test to the parameter $\theta = 0$. Korwar (1977) characterized the QB distribution in the context of the Rao-Rubin damage model. Consul (1990) discussed some properties and applications of the QB distribution, considered its moments and maximum likelihood estimation and applied it to several data sets. A good presentation and discussion of the QB distribution can be found in the book *Lagrangian Probability Distributions* by Consul and Famoye (2006).

According to Fazal (1976), a discrete random variable Y is said to follow the QB distribution if its pmf is given by

$$P(Y = y) = pq\binom{m}{y}(p + y\theta)^{y-1}[q + (m-y)\theta]^{m-y-1} / (1 + m\theta)^{m-1}, \quad y = 0,1,2,\ldots,m,$$

where $\binom{m}{y} = \dfrac{m!}{(m-y)!y!}$, $0 \le p < 1$, $q = 1 - p$, $0 \le \theta < 1$.

Remark 3. Note that for $\theta = 0$, we obtain the binomial distribution with parameters m and p as a particular case of quasi-binomial distribution.

The mean and the first factorial moment of the QB distribution are given by

$$E(Y) = mp \quad \text{and} \quad E\left[Y(Y-1)\right] = m(m-1)p - qp\sum_{j=0}^{m-2} \frac{m!\theta^j}{(m-j-2)!(1+m\theta)^{j+1}}.$$

Thus, the variance is given by

$$\mathrm{Var}(Y) = pq\left[m^2 - \sum_{j=0}^{m-2} \frac{m!\theta^j}{(m-j-2)!(1+m\theta)^{j+1}}\right].$$

Shenton (2006) gave an overview on the QB distribution and some properties were listed. The QB distribution converges in distribution to the GP distribution as m increases. The sum of quasi-binomial variates in general is not a quasi-binomial variate and properties of the binomial distribution, such as expressions of the partial sum of probabilities in integral form, are not readily available for the quasi-binomial distribution.

2.3. GPAR(1) process

Alzaid and Al-Osh (1993) introduced the GPAR(1) process and obtained some properties like mean, variance, autocorrelation function, conditional variance and conditional expectation. This process can be viewed as an extension of Poisson INAR(1) process by Al-Osh and Alzaid (1987). Al-Nachawati et al. (1997) compared two methods of estimating the parameters of the GPAR(1): Yule-Walker and Gaussian estimation. They observe that, for $\lambda > 1$, in small and larger sample sizes the Gaussian method has better results than YW estimator according the bias and MSE of estimates. For $\lambda \leq 1$, in small and larger sample sizes the YW and Gaussian methods presented similar results in terms of bias and MSE of estimates.

Alzaid and Al-Osh (1993) introduced the quasi-binomial operators as a sequence of random variables $\{S(n); n = 0,1,2,\ldots\}$ such that $S(n)$ has $\mathrm{QB}(p,\theta/\lambda,n)$ distribution for some p, θ and λ. Weiß (2008) called this operator as *"quasi-binomial thinning operator"*.

The quasi-binomial thinning operator has an interpretation in analogy with the binomial thinning operator. In the same way of the binomial thinning in the development of the Poisson INAR(1), the thinned variable $S(X)$ can be understood as the number of survivors to the next generation from a population with X individuals (Weiß, 2008). Unlike the assumption in the Poisson INAR(1) process that given $X_{t-1} = x$, the number of retained elements to time t is a random variable with binomial distribution $B(\alpha, x)$, in the development of the GPAR(1) process it was assumed that given $X_{t-1} = x$ the number of retained elements to time t has a quasi-binomial distribution $QB(p,\theta,x)$. An element of the Poisson process at time $t-1$, X_{t-1}, independently of the other elements and the time of the process has a constant probability of being retained to time t, in the GPAR(1) process the quasi-binomial thinning operator consider that the probability of retaining an element is not constant but might depend on the time and the number of elements already retained (Alzaid and Al-Osh, 1993).

Based on the following proposition, Alzaid and Al-Osh (1993) defined the first-order autoregressive process with generalized Poisson marginal distribution.

Proposition 1. *(Alzaid and Al-Osh, 1993) If X is distributed according to the $GP(\lambda,\theta)$ distribution and the quasi-binomial thinning is performed independently of X, then $S(X)$ is distributed according to $GP(p\lambda,\theta)$.*

The GPAR(1) process $\{X_t\}$ is defined as

$$X_t = S_t(X_{t-1}) + \epsilon_t, \qquad t \in \mathbb{N},$$

where $\{\epsilon_t\}$ is an i.i.d. sequence of random variables with $\mathrm{GP}(q\lambda,\theta)$ and $\{S_t(\cdot), t = 1,2,\ldots\}$ is a sequence of i.i.d. quasi-binomial operators independent of ϵ_t with $\mathrm{QB}(p,\theta/\lambda,\cdot)$ distribution, with $\lambda > 0$, $0 < \theta < 1$, $0 < p < 1$, $q = 1 - p$.

We have from the definition that $S_t(X_{t-1}) \mid (X_{t-1} = x)$ has $\mathrm{QB}(p,\theta/\lambda,x)$ distribution. Moreover, assuming that X_0 follows a $\mathrm{GP}(\lambda,\theta)$ distribution independent of $\{\epsilon_t\}$ and $\{S_t(\cdot)\}$, the GPAR(1) process is a strictly stationary Markov chain with generalized Poisson marginal distribution (Alzaid and Al-Osh, 1993).

The mean and variance of the GPAR(1) process are given in (2), the coefficients of skewness and kurtosis are given in (3). It is not difficult to verify that the conditional mean and variance of the GPAR(1) process given X_{t-1} are, respectively,

$$\mathrm{E}(X_t \mid X_{t-1} = x) = px + \mu_\epsilon$$

and

$$\mathrm{Var}(X_t \mid X_{t-1} = x) = pq\left[x^2 - \sum_{j=1}^{x-2} \frac{x!\theta^j}{(x-j-2)!(1+x\theta)^{j+1}}\right] + \sigma_\epsilon^2,$$

where $\mu_\epsilon = \mathrm{E}(\epsilon_t) = q\lambda/(1-\theta)$ and $\sigma_\epsilon^2 = \mathrm{Var}(\epsilon_t) = q\lambda/(1-\theta)^3$.

The autocorrelation function of the GPAR(1) is given by

$$\rho_X(k) = \mathrm{corr}(X_t, X_{t-k}) = p^{|k|}, \quad k \in \mathbb{Z}.$$

Note that the autocorrelation function only depends on the p parameter and that the its behaviour is similar to that of the AR(1). Another characteristic of the GPAR(1) process is that its variance is greater than the mean while the Poisson INAR(1) process has the mean equal to the variance.

2.4. GPDAR(1) process

Now, based on the quasi-binomial operator, we will define the new operator $R(\cdot)$ and, in the same way to Freeland (2010), the new first order autoregressive process with generalized Poisson difference distribution marginals. Our main proposal here is to extend the TINAR process of Freeland (2010), which is based upon the difference of two independent Poisson random variables, to a process that is based upon the difference of two independent generalized Poisson random variables. Our motivation for this is that the main feature of the GPDAR(1) process is more flexible in tails than the TINAR process. This is because the GPD distribution may have longer or shorter tails (Shahtahmassebi and Moyeed, 2014). The process admits positive and negative values and its sequence of innovations has a generalized Poisson difference distribution, as the marginal of the process.

Definition 2.1. *Let X and Y be two independent random variables with $GP(\lambda_1,\theta)$ and $GP(\lambda_2,\theta)$ distributions, respectively, $Z = X - Y$ following $GPD(\lambda_1,\lambda_2,\theta)$ distribution. We define the operator $R(\cdot)$ as*

$$R(Z) \mid Z \overset{d}{=} (S(X) - S(Y)) \mid (X - Y), \tag{4}$$

where $S(X) \mid (X = x)$ and $S(Y) \mid (Y = y)$ have $\mathrm{QB}(p, \theta/\lambda_1, x)$ and $\mathrm{QB}(p, \theta/\lambda_2, y)$ distributions, respectively, and the symbol "$U \overset{d}{=} V$" means that the random variables U and V have the same distribution.

We stress that, for $\theta = 0$, the quasi-binomial distribution becomes a binomial distribution in the same way as the generalized Poisson distribution becomes a Poisson distribution. Hence, the TINAR process of Freeland (2010) is indeed a particular case of our process, corresponding to $\theta = 0$.

From (4) we have that $R(Z) \overset{d}{=} S(X) - S(Y)$. Let X and Y be two independent random variables, so $S(X)$ and $S(Y)$ are independent random variables. From the definition of Alzaid and Al-Osh (1993) of the quasi-binomial operator we have that $S(X)$ and $S(Y)$ have GP distributions.

Let $\mathbb{N}_0 = \{0,1,2,\ldots\}$. Let $\{X_t\}_{t\in\mathbb{N}_0}$ and $\{Y_t\}_{t\in\mathbb{N}_0}$ be two independent GPAR(1) processes

$$X_t = S_t(X_{t-1}) + \epsilon_t, t \in \mathbb{N} \tag{5}$$

and

$$Y_t = S_t(Y_{t-1}) + \delta_t, t \in \mathbb{N}, \tag{6}$$

follow Generalized Poisson marginals with $\mathrm{GP}(\lambda_1, \theta)$ and $\mathrm{GP}(\lambda_2, \theta)$ distributions, respectively. Furthermore, the sequences $S_t(X_{t-1}) \mid (X_{t-1} = x)$ and $S_t(Y_{t-1}) \mid (Y_{t-1} = y)$ follow $\mathrm{QB}(p, \theta/\lambda_1, x), \mathrm{QB}(p, \theta/\lambda_2, y)$ distributions independent of $\{\epsilon_t\}_{t\in\mathbb{N}}$ and $\{\delta_t\}_{t\in\mathbb{N}}$, with $\lambda_1 > 0, \lambda_2 > 0, 0 < p < 1, 0 < \theta < 1$, $q = 1 - p$. The sequences $\{\epsilon_t\}_{t\in\mathbb{N}}$ and $\{\delta_t\}_{t\in\mathbb{N}}$ are independent with $\mathrm{GP}(q\lambda_1, \theta)$ and $\mathrm{GP}(q\lambda_2, \theta)$ distributions, respectively. The integer-valued first-order autoregressive process with generalized Poisson difference marginals is defined as follow.

Definition 2.2. *Let X_t and Y_t defined as above. We define the integer-valued first-order autoregressive process with generalized Poisson difference marginals, GPDAR(1), $\{Z_t\}_{t\in\mathbb{N}_0}$, as a sequence of a random variables such that*

$$Z_t \overset{d}{=} X_t - Y_t \overset{d}{=} R_t(Z_{t-1}) + \nu_t, \tag{7}$$

where $R_t(Z_{t-1})$ is a sequence of operators defined as (4) and $\nu_t = \epsilon_t - \delta_t$.

We have from (5) and (6) that ϵ_t and δ_t follow $\mathrm{GP}(q\lambda_1, \theta)$ and $\mathrm{GP}(q\lambda_2, \theta)$ distributions, respectively, so we have that ν_t follows the generalized Poisson difference distribution $\mathrm{GPD}(q\lambda_1, q\lambda_2, \theta)$.

Remark 4. This process can be viewed as an extension of the PDINAR(1) proposed by Alzaid and Omair (2014).

3. Properties of the process

In this section, we shall present some properties of the GPDAR(1) process, such as mean, variance, skewness and kurtosis coefficients, probability generating and cumulant generating functions. The autocorrelation and spectral density functions are also obtained. Follow, we present the marginal distribution properties of the GPDAR(1) process and the distribution properties of the sequence $\{\nu_t\}$ can be obtained in the same way of it.

The mean and variance of Z_t are given, respectively, by

$$\mu_Z = \mathrm{E}(Z_t) = \mathrm{E}(X_t) - \mathrm{E}(Y_t) = \frac{\lambda_1 - \lambda_2}{1 - \theta} \tag{8}$$

and

$$\sigma_Z^2 = \mathrm{Var}(Z_t) = \mathrm{Var}(X_t) + \mathrm{Var}(Y_t) = \frac{\lambda_1 + \lambda_2}{(1-\theta)^3}. \tag{9}$$

Remark 5. Note that $|\mu Z| < \sigma_Z^2$, then the GPDAR(1) process can be used as a model for overdispersed integer valued time series.

Consul (1986) considered the difference of two independent random variables with generalized Poisson distribution and obtained some properties. From those we have that the pgf of Z_t is given by

$$G_Z(s) = G_X(s) \cdot G_Y(s) = e^{[\lambda_1(t_1-1)+\lambda_2(t_2-1)]},$$

where $G_X(s)$ and $G_Y(s)$ are the X_t and $-Y_t$ pgf's and $t_1 = se^{\theta(t_1-1)}$, $t_2 = s^{-1}e^{\theta(t_2-1)}$.

The cumulant generating function (cgf) of Z_t is given by

$$\psi(\beta) = \lambda_1(e^{s_1} - 1) + \lambda_2(e^{s_2} - 1) = (s_1 - \beta)\lambda_1 / \theta + (s_2 + \beta)\lambda_2 / \theta,$$

where $s_1 = \beta + \theta(e^{s_1} - 1)$, $s_2 = -\beta + \theta(e^{s_2} - 1)$, and the first four cumulants are

$$k_1 = \frac{\lambda_1 - \lambda_2}{1-\theta}, \qquad k_2 = \frac{\lambda_1 + \lambda_2}{(1-\theta)^3},$$

$$k_3 = \frac{(\lambda_1 - \lambda_2)(1+2\theta)}{(1-\theta)^5}, \quad k_4 = \frac{(\lambda_1 + \lambda_2)(1+8\theta+6\theta^2)}{(1-\theta)^7}.$$

The coefficients of skewness and kurtosis of Z_t are given by

$$\gamma_1 = \frac{(\lambda_1 - \lambda_2)(1+2\theta)}{\sqrt{(\lambda_1+\lambda_2)^3}\sqrt{(1-\theta)}} \quad \text{and} \quad \gamma_2 = 3 + \frac{1+8\theta+6\theta^2}{(\lambda_1+\lambda_2)(1-\theta)},$$

respectively.

Remark 6. Note that if $\lambda_1 = \lambda_2$, we have a symmetric process with zero mean.

It is possible to show that the autocovariance function of Z_t process is given by

$$\gamma(h) = \text{cov}(Z_t, Z_{t-h}) = \frac{\lambda_1 + \lambda_2}{(1-\theta)^3} p^{|h|}, h \in \mathbb{Z},$$

and the autocorrelation function (ACF) at lag h is given by

$$\rho(h) = \text{Corr}(Z_t, Z_{t-h}) = p^{|h|}, \quad h \in \mathbb{Z}.$$

Since $0 < p < 1$, the autocorrelation function decays exponentially, like in the AR(1)process.

Remark 7. The process defined in (7) has positive autocorrelation. We can define a GPDAR(1) process with negative autocorrelation. In the same way of Freeland (2010), we define

$$Z_t = \begin{cases} X_t - Y_t, & t = 0, 2, 4, \ldots \\ Y_t - X_t, & t = 1, 3, 5, \ldots \end{cases}$$

Then, we have in this case that $\rho(h) = \text{Corr}(Z_t, Z_{t-h}) = (-p)^h$, for $h \in \mathbb{N}$. The main properties of this process can be obtained as in the process defined in (7).

The spectral density function $f(\omega)$ of the GPDAR(1) process is

$$f(\omega) = \frac{(\lambda_1 + \lambda_2)}{2\pi(1-\theta)^3} \frac{2p(\cos(\omega) - p)}{(1 + p^2 - 2p\cos(\omega))}, \quad \omega \in (-\pi, \pi].$$

Figure 6 displays a sample autocorrelation based on 100 simulated values and the spectral density function for $p = 0.3, \lambda_1 = 1, \lambda_2 = 2$ and $\theta = 0.8$.

Now, we present two results about the r.v. $R_t(Z_{t-1})$ and the GPDAR(1) process. In Proposition 2, we obtain the conditional probability function of the r.v. $R_t(Z_{t-1})$ given Z_{t-1}, this result is particularly useful if our interest is to obtain the transition probabilities from the process. In Proposition 3, the conditional expectation and variance of GPDAR(1) process are obtained. The conditional expectation expression is directly linked to 1-step ahead forecasts and used the conditional least squares estimators for unknown parameters of the process. The proofs of the propositions can be found in the Appendix.

Proposition 2. The conditional probability function of the random variable $R_t(Z_{t-1})$ given Z_{t-1} is given by

$$\Pr(R_t(Z_{t-1}) = j \mid Z_{t-1} = k) = \sum_{l=0}^{\infty} \sum_{i=0}^{\min(l,l+k-j)} \frac{(\lambda_1,\theta)_{k+l}(\lambda_2,\theta)_l \mathrm{e}^{-2l\theta} f(i+j;p,\theta/\lambda_1,k+l) f(i;p,\theta/\lambda_2,l)}{\Sigma_{y=0}^{\infty}(\lambda_2,\theta)_y(\lambda_1,\theta)_{y+k}\mathrm{e}^{-2y\theta}},$$

for $j \geq 0, k > 0$,and

$$\Pr(R_t(Z_{t-1}) = j \mid Z_{t-1} = k) = \sum_{l=0}^{\infty} \sum_{i=0}^{\min(l-j,l-k)} \frac{(\lambda_1,\theta)_l(\lambda_2,\theta)_{l-k} \mathrm{e}^{-2l\theta} f(i;p,\theta/\lambda_1,l) f(i+j;p,\theta/\lambda_2,l-k)}{\sum_{y=0}^{\infty}(\lambda_2,\theta)_y(\lambda_1,\theta)_{y+k}\mathrm{e}^{-2(y+k)\theta}},$$

for $j \geq 0, k < 0$, where $(\lambda,\theta)_y = \dfrac{\lambda(\lambda+y\theta)^{y-1}}{y^!}$ and $f(i;p,\theta,n)$ is the QB's probability mass function with parameters (p,θ,n).

Proposition 3 Let $\{Zt\}_{t\epsilon\mathrm{N}0}$ be a GPDAR(1) process according to Definition 2.2. The conditional expectation of Zt given $Zt-1$ is given by

$$\mathrm{E}(Z_t \mid Z_{t-1}) = pZ_{t-1} + \mu_v,$$

where $\mu_v = E(vt) = E(\epsilon t) - \mathrm{E}(\delta_t) = \dfrac{q(\lambda_1 - \lambda_2)}{1-0}$ and the conditional variance is given by

$$\mathrm{Var}(Z_t \mid Z_{t-1} = z) = \sum_{y=0}^{\infty} \frac{(\lambda_2,\theta)_y(\lambda_1,\theta)_{y+z}\{p[(z+y)(z+y(1-p)) + y^2 - q[W_{(z+y)} + W_{(y)}]\}\mathrm{e}^{-2y\theta}}{\Sigma_{y=0}^{\infty}(\lambda_2,\theta)_y(\lambda_1,\theta)_{z+y}\mathrm{e}^{-2y\theta}}$$

$-(pz+\mu_v)^2$, for $z \geq 0$, and

$$\mathrm{Var}(Z_t \mid Z_{t-1} = z) = \sum_{x=0}^{\infty} \frac{(\lambda_2,\theta)_{(x-z)}(\lambda_1,\theta)_x\{p[(x-z)(x(1-p)-z) + x^2 - q(W_{(x)} + W_{(x-z)})]\}\mathrm{e}^{-2x\theta}}{\Sigma_{y=0}^{\infty}(\lambda_2,\theta)_y(\lambda_1,\theta)_{y+z}\mathrm{e}^{-2(y+z)\theta}}$$

$-(pz+\mu_v)^2$, for $z < 0$. Where $w_{(x)} = \sum_{j=0}^{x-1} \dfrac{x!\theta^j}{(x-j-2)!(1+x\theta)^{j+1}}$ and $(\lambda,\theta)_y$ was defined as in Proposition

Remark 8. The expressions of the conditional variance are not analytically simple, but it is not difficult to evaluate them numerically.

4. Estimation

In this section, we outline the estimation of the unknown parameters of the GPDAR(1) process. We investigate the estimation method of moments with Yule-Walker .

Considering the skew process, $\lambda_1 \neq \lambda_2$, we can use Yule-Walker estimation to estimate p. From a sample $Z_1, Z_2, \ldots, Z_T$, based on the fact that the correlation function is given by $\rho(h) = p^{|h|}$ we consider the Yule-Walker estimator given by

$$\hat{p} = \hat{\rho}(1) = \frac{\Sigma_{t=1}^{T-1}(Z_t - \bar{Z})(Z_{t+1} - \bar{Z})}{\Sigma_{t=1}^{T}(Z_t - \bar{Z})^2}.$$

Based on the method of moments, using the sample mean and variance $\bar{Z} = \frac{1}{T}\sum_{t=1}^{T} Z_t$, $\hat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}(Z_t - \bar{Z})^2$, and the two first moments given by (8) and (9), we have

$$\bar{Z} = \frac{\hat{\lambda}_1 - \hat{\lambda}_2}{1 - \hat{\theta}} \text{ and } \hat{\sigma}^2 = \frac{\hat{\lambda}_1 + \hat{\lambda}_2}{(1 - \hat{\theta})^3}.$$

Then, after a simple algebra, the estimators for λ_1 and λ_2 are given by

$$\hat{\lambda}_1 = \frac{1}{2}[\hat{\sigma}^2(1 - \hat{\theta})^3 + \bar{Z}(1 - \hat{\theta})]$$

and

$$\hat{\lambda}_2 = \frac{1}{2}[\hat{\sigma}^2(1 - \hat{\theta})^3 - \bar{Z}(1 - \hat{\theta})].$$

Using the sample third moment $m_3 = \frac{1}{T}\sum_{t=1}^{T}(Z_t - \bar{Z})^3$ and the third cumulant, the moment estimator of θ satisfies the equation

$$(1 - \hat{\theta})^4 m_3 - \bar{Z}(1 + 2\hat{\theta}) = 0. \tag{10}$$

It is not difficult to show that, if m_3 and $\bar{Z}$ are both positive or negative, Equation (10) has a solution in $(0,1)$ and one can find this solution numerically using some computer software, for example, the R software (see http://www.r-project.org). Note that the moment estimators for λ_1 and λ_2 are well defined if $\hat{\sigma}^2(1 - \hat{\theta})^2 > |\bar{Z}|$. Furthermore, if $\lambda_1 = \lambda_2$, we have a symmetric process with zero mean, so the estimators presented above do not have sense in this case and is necessary an estimator more adequate.

Consider the symmetric process with $\lambda_1 = \lambda_2 = \lambda$. Using the sample variance $\hat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}(Z_t - \bar{Z})^2$ and the variance of the process, $\sigma_Z^2 = 2\lambda/(1 - \theta)^3$, the moment estimator for λ is given by

$$\hat{\lambda} = \frac{1}{2}\hat{\sigma}^2(1-\hat{\theta})^3.$$

Now, consider the fourth cumulant k_4 and variance σ^2, defined in Section 3, and the well known relation $k_4 = \mu_4 - 3\mu_2^2$, where μ_4 and μ_2 are the fourth and second central moments. Using the second and fourth sample central moment, $m_2 = \frac{1}{T}\sum_{t=1}^{T}(Z_t - \bar{Z})^2$ and $m_4 = \frac{1}{T}\sum_{t=1}^{T}(Z_t - \bar{Z})^4$, respectively, the moment estimator of θ satisfies

$$(1-\hat{\theta})^4\hat{k}_4 - \hat{\sigma}^2(1+8\hat{\theta}+6\hat{\theta}^2) = 0, \tag{11}$$

where $\hat{k}_4 = m_4 - 3m_2^2$. The Equation (11) has solution since that $\hat{k}_4 > \hat{\sigma}^2$. The moment estimator of θ do not have closed form but we can find it numerically as in the last case.

The conditional least squares method consists of minimizing the function

$$Q_T(\beta) = \sum_{t=2}^{T}[Z_t - E(Z_t \mid Z_{t-1})]^2,$$

where β is a vector of unknown parameters of the process. Here, the function to be minimized is given by

$$Q_T(p,\mu_v) = \sum_{t=2}^{T}(Z_t - pZ_{t-1} - \mu_v)^2 = \sum_{t=2}^{T}[Z_t - pZ_{t-1} - (1-p)\mu]^2.$$

Note that using this method we can not obtain estimators for λ_1, λ_2 and θ, but only p and $\mu = (\lambda_1 - \lambda_2)/(1-\theta)$, for this reason, this method of estimation was not considered.

5. study of simulation

A Monte Carlo simulation experiment was performed to study the behaviour of the Yule-Walker estimators. The number of Monte Carlo replications was 5000 and the considered sample sizes were T = 50, 100 and 200. For the values of parameters, we considered $p = 0.5, 0.6, \lambda_1 = 1,2, \lambda_2 = 1,2$ and $\theta = 0.6$. The Monte Carlo simulation was performed using the R programming language, see http://www.r-project.org.Table 1 presents the bias and mean squared error (MSE) of the estimates of the parameters of the GPDAR(1) process. From the results we conclude that estimates of the parameters are reasonable, there are convergence to the true parameters values, and the bias and MSE decrease as the size sample increases as expected. The bias and MSE of λ_1 and λ_2 estimates decreases non-uniformly. Note that the MSE values of λ_1 and λ_2 estimators are large for $\theta = 0.6$, this is justified because the random sample will have a large variance and this estimators depends on it directly.

6. Application to real data set

In this section, we present an application of the GPDAR(1) model to the GLOBAL Land-Ocean temperature index. We consider the monthly GLOBAL Land-Ocean temperature index in 0.01 degrees Celsius for 1950-1962 update december 2015. The temperature index in 0.01 degree Celsius belongs to $\mathbb{Z}$ and is our time series of interest here in this section. The data set consist of 156 observations. The data has been downloaded from the National Oceanic and Atmospheric Administration (NOAA), U.S. Department of Commerce, Earth System Research Laboratory (http://www.esrl.noaa.gov/psd/data/climateindices/list), originally from http://data.giss.nasa.gov/gistemp/tabledata_v3/GLB.Ts.txt.

Table 2 provides some descriptive measures for the temperature index, which include central tendency statistics, variance, skewness and kurtosis, among others. We see that the data set assumes positive and negative integer values.

The time series data and their sample autocorrelations and partial autocorrelations are displayed in Figure 7. Analyzing Figure 7, we conclude that a first order autoregressive model may be appropriate for the given data series, because of the clear cut-off after lag 1 in the partial autocorrelations. Furthermore, the behavior of the series indicates that it may be mean stationary. We compare our model with the TINAR(1) process introduced by Freeland (2010) (asymmetric version). The TINAR(1) process considered here has marginals following a Skellam distribution with parameters $\lambda_1(1-p)^{-1}$ and $\lambda_2(1-p)^{-1}$, i.e., the marginals are distributed as $Y_1 - Y_2$, where Y_1 and Y_2 are two independent Poisson random variables with mean $\lambda_1(1-p)^{-1}$ and $\lambda_2(1-p)^{-1}$, respectively, and p is the associated thinning parameter of this process.

In Table 3 we present the estimates of the parameters with their MdAE (median absolute error) and estimated quantities (index of dispersion, skewness and kurtosis) for the fitted models. The MdAE statistic is defined as follows. For $t = 2,\ldots,T$, consider the expected value of the observation at the previous time, $E[Z_t \mid Z_{t-1}] = pZ_{t-1} + (1-p)(\lambda_1 - \lambda_2)/(1-\theta)$ (for GPDAR(1) process). The MdA is obtained as the median of the $|Z_t - E(Z_t \mid Z_{t-1})|$. In general it is expected that the best model to fit data presents the smallest values for this statistic. From Table 3, note that all two models exhibit good fit of mean and variance, but only the GPDAR(1) model give a reasonable fit of skewness and kurtosis, i.e., the TINAR(1) model is not able to reproduce the skewness and kurtosis. Furthermore, the values of MdAE statistic indicate that the GPDAR(1) model showed better result compared with TINAR(1) model. Based on these facts, we conclude that the GPDAR(1) model capture more information of these data. The sample autocorrelations of the residuals obtained from GPDAR(1) model are shown in Figure 8. From this figure, we conclude that the proposed model fitted the data well.

7. Concluding remarks

In this paper, we introduced the GPDAR(1) process. The main advantages of this new model is the skewness, possible negative values for the time series and the distribution of the difference of two GP random variables can have longer or shorter tails compared to the PD distribution and the GPD distribution can be viewed as a generalisation of the PD distribution. We did a review of the GP, QB distribution and the GPAR(1) process. The properties of the GPDAR(1) process was presented, mean, variance, skewness and kurtosis coefficients, probability generating and cumulant functions. The autocorrelation and spectral density are obtained. The conditional expectation and variance were derived. A Monte Carlo simulation study shows that the Yule-Walker estimators have a reasonable behaviour. An application to real data set shows that the GPDAR(1) process is adequate to fit skew data set with positive and negative values.

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Appendix

Proof of Proposition 2. First, let us consider the case $j \geq 0$ and $k \geq 0$. We have,

$$P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid Z_{t-1} = k) = \frac{P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j, Z_{t-1} = k)}{P(Z_{t-1} = k)}$$

$$= \frac{1}{P(Z_{t-1} = k)} \sum_{l=0}^{\infty} P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid X_{t-1} = k+l, Y_{t-1} = l) P(X_{t-1} = k+l, Y_{t-1} = l).$$

We know that the random variables X_{t-1} and Y_{t-1} have $GP(\lambda_1, \theta)$ and $GP(\lambda_2, \theta)$ distributions, respectively. So, we find $P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid X_{t-1} = k+l, Y_{t-1} = l)$. Define $U = S_t(X_{t-1}) \mid X_{t-1} = k+l$ and $V = S_t(Y_{t-1}) \mid Y_{t-1} = l$, with $QB(p, \theta/\lambda_1, k+l)$ and $QB(p, \theta/\lambda_2, l)$ distributions respectively. X_t and Y_t are independent random variables, so the conditional distribution $P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid X_{t-1} = k+l, Y_{t-1} = l)$ is the convolution of two random variables with QB distributions.

Thus, let $*$ denote convolution, let $f(i; p, \theta, n)$ be the QB's pmf, this is

$$f(i; p, \theta, n) = \frac{pq \binom{n}{i} (p + y\theta)^{y-1} [q + (n-y)\theta]^{n-y-1}}{(1+n\theta)^{n-1}}.$$

Then,

$$P(U - V = j) = \sum_i f(i+j; p, \theta/\lambda_1, k+l) f(i; p, \theta/\lambda_2, l).$$

We have

$$0 \leq i \leq l \text{ and } i \leq k+l-j \Rightarrow 0 \leq i \leq \min(l, l+k-j).$$

Thus,

$$P(U - V = j) = \sum_{i=0}^{\min(l, l+k-j)} f(i+j; p, \theta/\lambda_1, k+l) f(i; p, \theta/\lambda_2, l)$$

and

$$P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid Z_{t-1} = k)$$

$$= \frac{1}{P(Z_{t-1} = k)} \sum_{l=0}^{\infty} P(S_t(X_{t-1}) - S_t(Y_{t-1}) = j \mid X_{t-1} = k+l, Y_{t-1} = l) P(X_{t-1} = k+l, Y_{t-1} = l)$$

$$= \sum_{l=0}^{\infty} \frac{P(X_{t-1} = k+l, Y_{t-1} = l)}{P(Z_{t-1} = k)} \sum_{i=0}^{\min(l, l+k-j)} f(i+j; p, \theta/\lambda_1, k+l) f(i; p, \theta/\lambda_2, l)$$

$$= \frac{\Sigma_{l=0}^{\infty} \Sigma_{i=0}^{\min(l, l+k-j)} (\lambda_1, \theta)_{k+l} (\lambda_2, \theta)_l e^{-2l\theta} f(i+j; p, \theta/\lambda_1, k+l) f(i; p, \theta/\lambda_2, l)}{\Sigma_{y=0}^{\infty} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+k} e^{-2y\theta}}.$$

For $j \geq 0, k < 0$ we have

$$P(S_t(X_{t-1})-S_t(Y_{t-1})=j \mid Z_{t-1}=k)=\frac{P(S_t(X_{t-1})-S_t(Y_{t-1})=j, Z_{t-1}=k)}{P(Z_{t-1}=k)}$$

$$=\frac{1}{P(Z_{t-1}=k)}\sum_{l=0}^{\infty} P(S_t(X_{t-1})-S_t(Y_{t-1})=j \mid X_{t-1}=l, Y_{t-1}=l-k)P(X_{t-1}=l, Y_{t-1}=l-k).$$

In this case, we have

$$i \le l-j \text{ and } 0 \le i \le l-k \Rightarrow 0 \le i \le \min(l-j, l-k)$$

and

$$P(S_t(X_{t-1})-S_t(Y_{t-1})=j \mid Z_{t-1}=k)$$

$$=\frac{1}{P(Z_{t-1}=k)}\sum_{l=0}^{\infty} P(S_t(X_{t-1})-S_t(Y_{t-1})=j \mid X_{t-1}=l, Y_{t-1}=l-k)P(X_{t-1}=l, Y_{t-1}=l-k)$$
$$=\sum_{l=0}^{\infty}\frac{P(X_{t-1}=l, Y_{t-1}=l-k)}{P(Z_{t-1}=k)}\sum_{i=0}^{\min(l-j,l-k)} f(i+j; p, \theta/\lambda_1, l) f(i; p, \theta/\lambda_2, l-k)$$
$$=\frac{\Sigma_{l=0}^{\infty}\Sigma_{i=0}^{\min(l-j,l-k)}(\lambda_1,\theta)_l(\lambda_2,\theta)_{l-k}e^{-2l\theta} f(i+j; p, \theta/\lambda_1, l) f(i; p, \theta/\lambda_2, l-k)}{\Sigma_{y=0}^{\infty}(\lambda_2,\theta)_y(\lambda_1,\theta)_{y+k}e^{-2(y+k)\theta}}.$$

Proof of Proposition 3. Using the definition of Z_t, we have

$$E(Z_t \mid Z_{t-1}=z)=E(S_t(X_{t-1})-S_t(Y_{t-1})+\nu_t \mid X_{t-1}-Y_{t-1}=z)$$
$$=E(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}-Y_{t-1}=z)+E(\nu_t)$$

since that ν_t, X_{t-1} and Y_{t-1} are independent random variables and

$$E(\nu_t)=E(\epsilon_t)-E(\delta_t)=\frac{q\lambda_1}{1-\theta}-\frac{q\lambda_2}{1-\theta}=\frac{q(\lambda_1-\lambda_2)}{1-\theta},$$

to find the conditional expectation of the Z_t given $Z_{t-1}=z$ we have find

$$E(S_t(X_{t-1})+S_t(Y_{t-1}) \mid X_{t-1}-Y_{t-1}=z).$$

For $z \ge 0$ we have,

$$E(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}-Y_{t-1}=z)$$
$$=\sum_{r=-\infty}^{\infty} rP(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}-Y_{t-1}=z)$$
$$=\sum_{r=-\infty}^{\infty}\sum_{y=0}^{\infty} rP(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}=z+y, Y_{t-1}=y)$$
$$=\sum_{r=-\infty}^{\infty}\sum_{y=0}^{\infty} r\frac{P(S_t(X_{t-1})-S_t(Y_{t-1}), X_{t-1}=z+y, Y_{t-1}=y)}{P(X_{t-1}-Y_{t-1}=z)}$$

$$=\frac{\Sigma_{y=0}^{\infty}P(X_{t-1}=z+y)P(Y_{t-1}=y)}{P(X_{t-1}-Y_{t-1}=z)}\sum_{r=-\infty}^{\infty} rP(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}=z+y, Y_{t-1}=y)$$
$$=\frac{\Sigma_{y=0}^{\infty}P(X_{t-1}=z+y)P(Y_{t-1}=y)}{P(X_{t-1}-Y_{t-1}=z)}E(S_t(X_{t-1})-S_t(Y_{t-1}) \mid X_{t-1}=z+y, Y_{t-1}=y).$$

As defined in 2, we know that the random variables $S_t(X_{t-1}) \mid X_{t-1}=z+y$ and

$S_t(Y_{t-1}) \mid Y_{t-1} = y$ have $QB(p, \theta/\lambda_1, z+y)$ and $QB(p, \theta/\lambda_2, y)$ distributions, respectively. Then,

$$E(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y) = (z+y)p - yp = zp$$

Then,

$$E(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} - Y_{t-1} = z) = \frac{\Sigma_{y=0}^{\infty} P(X_{t-1} = z+y) P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} zp = zp.$$

For $z < 0$, we have

$$\begin{aligned}
&E(S_t(X_{t-1}) + S_t(Y_{t-1}) \mid X_{t-1} - Y_{t-1} = z) \\
&= \sum_{r=-\infty}^{\infty} r P(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} - Y_{t-1} = z) \\
&= \sum_{r=-\infty}^{\infty} \sum_{x=0}^{\infty} r P(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x-z) \\
&= \sum_{r=-\infty}^{\infty} \sum_{x=0}^{\infty} r \frac{P(S_t(X_{t-1}) - S_t(Y_{t-1}), X_{t-1} = x, Y_{t-1} = x-z)}{P(X_{t-1} - Y_{t-1} = z)}
\end{aligned}$$

$$= \frac{\Sigma_{x=0}^{\infty} P(X_{t-1} = z+y) P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} \sum_{r=-\infty}^{\infty} r P(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x-z)$$

$$= \frac{\Sigma_{x=0}^{\infty} P(X_{t-1} = x) P(Y_{t-1} = x-z)}{P(X_{t-1} - Y_{t-1} = z)} E(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x-z),$$

so, as above,

$$E(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x-z) = xp - (x-z)p = zp.$$

Then,

$$E(S_t(X_{t-1}) - S_t(Y_{t-1}) \mid X_{t-1} - Y_{t-1} = z) = zp \frac{\Sigma_{x=0}^{\infty} P(X_{t-1} = x) P(Y_{t-1} = x-z)}{P(X_{t-1} - Y_{t-1} = z)} = zp.$$

Then, for $Z_{t-1} = z, z \in \mathbb{Z}$, we have

$$E(Z_t \mid Z_{t-1}) = pZ_{t-1} + \mu_v,$$

where $\mu_v = E(v_t)$.

To find conditional variance, $\mathrm{Var}(Z_t \mid Z_{t-1} = z) = \mathrm{E}(Z_t^2 \mid Z_{t-1} = z) - [\mathrm{E}(Z_t \mid Z_{t-1} = z)]^2$, we need to find $\mathrm{E}(Z_t^2 \mid Z_{t-1} = z)$. In the same way of conditional expectation, for $z \geq 0$ we have

$$E(Z_t^2 \mid Z_{t-1} = z) = E[(S_t(X_{t-1}) - S_t(Y_{t-1}))^2 \mid X_{t-1} - Y_{t-1} = z]$$

$$= \sum_{y=0}^{\infty} \frac{P(X_{t-1} = z+y) P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} E[((S_t(X_{t-1}) - S_t(Y_{t-1}))^2 \mid X_{t-1} = z+y, Y_{t-1} = y)], \tag{12}$$

where,

$$\mathrm{E}[(S_t(X_{t-1}) - S_t(Y_{t-1})^2 \mid X_{t-1} = z+y, Y_{t-1} = y)]$$

$$= \mathrm{E}(S_t^2(X_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y)) + E(S_t^2(Y_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y))$$
$$-\mathrm{E}[S_t(X_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y)]\mathrm{E}[S_t(Y_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y)].$$

We know that if the random variable U has quasi-binomial distribution then $\mathrm{Var}(U)$ is given by (2), so

$$E(S_t^2(X_{t-1}) \mid X_{t-1} = z+y) = \mathrm{Var}(S_t(X_{t-1}) \mid X_{t-1}) + E^2(S_t(X_{t-1}) \mid X_{t-1})$$

$$= pq\left[(y+z)^2 - \sum_{j=0}^{z+y-1} \frac{(z+y)!\theta^j}{(z+y-j-2)!(1+(z+y)\theta)^{j+1}}\right] + (z+y)^2 p^2$$

$$= pq(y+z)^2 - pqW_{(z+y)} + p^2(z+y)^2 = p[(z+y)^2 - qW_{(z+y)}] \tag{13}$$

and

$$\mathrm{E}(S_t^2(Y_{t-1}) \mid Y_{t-1} = y) = p(y^2 - qW_y), \tag{14}$$

but $W_{(x)} = \sum_{j=0}^{x-1} \frac{x!\theta^j}{(x-j-2)!(1+x\theta)^{j+1}}$. X_{t-1} and Y_{t-1} are independent random variables, therefore

$$\mathrm{E}[S_t(X_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y)]\mathrm{E}[S_t(Y_{t-1}) \mid X_{t-1} = z+y, Y_{t-1} = y)]$$
$$= [p(z+y)][py] = y(z+y)p^2. \tag{15}$$

Replacing (13), (14) and (15) in (12). We have

$$E(Z_t^2 \mid Z_{t-1} = z)$$

$$= \sum_{y=0}^{\infty} \frac{P(X_{t-1} = z+y)P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} \{p[(z+y)^2 - qW_{(z+y)}] + p(y^2 - qW_{(y)}) - y(z+y)p^2\}$$

$$= \sum_{y=0}^{\infty} \frac{P(X_{t-1} = z+y)P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} \{p[(z+y)(z+y(1-p)) + y^2] - pq[W_{(z+y)} - W_{(y)}]\}.$$

Also,

$$\frac{P(X_{t-1} = z+y)P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} = \frac{\mathrm{e}^{-\lambda_1 - \lambda_2 - z\theta - 2y\theta}(\lambda_2, \theta)_y (\lambda_1, \theta)_{z+y}}{\Sigma_{y=0}^{\infty} \mathrm{e}^{-\lambda_1 - \lambda_2 - z\theta - 2y\theta}(\lambda_2, \theta)_y (\lambda_1, \theta)_{z+y}},$$

where $(\lambda, \theta)_x = \frac{\lambda(\lambda + x\theta)^{x-1}}{x!}$.

Thus,

$$\mathrm{E}(Z_t^2 \mid Z_{t-1} = z)$$

$$= \sum_{y=0}^{\infty} \frac{P(X_{t-1} = z+y)P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} \{p[(z+y)(z+y(1-p)) + y^2] - pq[W_{(z+y)} - W_{(y)}]\}$$

$$= \frac{\Sigma_{y=0}^{\infty} e^{-2y\theta}(\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z} \{p[(z+y)(z+y(1-p)) + y^2] - pq[W_{(z+y)} - W_{(y)}]\}}{\Sigma_{y=0}^{\infty} e^{-2y\theta}(\lambda_2, \theta)_y (\lambda_1, \theta)_{z+y}}.$$

Then, the conditional variance given $Z_{t-1} = z, z \geq 0$ is

$$\mathrm{Var}(Z_t \mid Z_{t-1} = z) = \mathrm{E}(Z_t^2 \mid Z_{t-1} = z) - [\mathrm{E}(Z_t \mid Z_{t-1} = z)]^2$$
$$= \frac{\Sigma_{y=0}^{\infty} e^{-2y\theta} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z} \{p[(z+y)(z+y(1-p)) + y^2 - q[W_{(z+y)} + W_{(y)}]\}}{\Sigma_{y=0}^{\infty} e^{-2y\theta} (\lambda_2, \theta)_y (\lambda_1, \theta)_{z+y}}$$
$$-(pZ_{t-1} + \mu_v)^2.$$

For $z < 0$ we have

$$\mathrm{E}(Z_t^2 \mid Z_{t-1} = z) = \mathrm{E}[(S_t(X_{t-1}) - S_t(Y_{t-1}))^2 \mid X_{t-1} - Y_{t-1} = z]$$
$$= \sum_{x=0}^{\infty} \frac{P(X_{t-1} = x) P(Y_{t-1} = x - z)}{P(X_{t-1} - Y_{t-1} = z)} \mathrm{E}[((S_t(X_{t-1}) - S_t(Y_{t-1}))^2 \mid X_{t-1} = x, Y_{t-1} = x - z)].$$

But

$$\mathrm{E}[(S_t(X_{t-1}) - S_t(Y_{t-1})^2 \mid X_{t-1} = x, Y_{t-1} = x - z)]$$
$$= \mathrm{E}(S_t^2(X_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z)) + \mathrm{E}(S_t^2(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z))$$
$$-\mathrm{E}[S_t(X_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z)]\mathrm{E}[S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z)]. \tag{16}$$

We know that if the random variable U has quasi-binomial distribution then $\mathrm{Var}(U)$ is given by (2), so

$$E(S_t^2(X_{t-1}) \mid X_{t-1} = x) = \mathrm{Var}(S_t(X_{t-1}) \mid X_{t-1} = x) + E^2(S_t(X_{t-1}) \mid X_{t-1} = x)$$
$$= p[x^2 - qW_{(x)}] \tag{17}$$

and

$$E(S_t^2(Y_{t-1}) \mid Y_{t-1} = x - z) = p[(x-z)^2 - qW_{(x-z)}], \tag{18}$$

where $W_{(x)} = \sum_{j=0}^{x-2} \frac{x!\theta^j}{(x-j-2)!(1+x\theta)^{j+1}}$. X_{t-1} and Y_{t-1} are independent random variable, therefore

$$\mathrm{E}[S_t(X_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z)]\mathrm{E}[S_t(Y_{t-1}) \mid X_{t-1} = x, Y_{t-1} = x - z)]$$
$$= x(x-z)p^2. \tag{19}$$

Replacing (17), (18) and (19) in (16), we obtain

$$\mathrm{E}(Z_t^2 \mid Z_{t-1} = z)$$
$$= \sum_{x=0}^{\infty} \frac{P(X_{t-1} = x) P(Y_{t-1} = x - z)}{P(X_{t-1} - Y_{t-1} = z)} \{p[x^2 - qW_{(x)}] + p[(x-z)^2 - qW_{(x-z)}] - x(x-z)p^2\}$$
$$= \sum_{x=0}^{\infty} \frac{P(X_{t-1} = x) P(Y_{t-1} = x - z)}{P(X_{t-1} - Y_{t-1} = z)} \{p[(x-z)(x(1-p) - z) + x^2 - q(W_{(x)} + W_{(x-z)})]\}.$$

We have that

$$\frac{P(X_{t-1} = x) P(Y_{t-1} = x - z)}{P(X_{t-1} - Y_{t-1} = z)} = \frac{\mathrm{e}^{-\lambda_1 - \lambda_2 + z\theta - 2x\theta} (\lambda_2, \theta)_{(x-z)} (\lambda_1, \theta)_x}{\Sigma_{y=0}^{\infty} \mathrm{e}^{-\lambda_1 - \lambda_2 - z\theta - 2y\theta} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z}}$$

where $(\lambda, \theta)_x = \frac{\lambda(\lambda + x\theta)^{x-1}}{x!}$. Then,

$$\mathrm{E}(Z_t^2 \mid Z_{t-1} = z)$$

$$= \sum_{x=0}^{\infty} \frac{P(X_{t-1} = z + y)P(Y_{t-1} = y)}{P(X_{t-1} - Y_{t-1} = z)} \{p[(x)^2 - qW_{(x)}] + p((x-z)^2 - qW_{(x-z)}) - x(x-z)p^2\}$$

$$= \frac{\Sigma_{y=0}^{\infty} \mathrm{e}^{-2x\theta} (\lambda_2, \theta)_{(x-z)} (\lambda_1, \theta)_x \{p[(x-z)(x(1-p)-z) + x^2 - q(W_{(x)} + W_{(x-z)})]\}}{\Sigma_{y=0}^{\infty} \mathrm{e}^{-2(y+z)\theta} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z}}$$

Thus, the conditional variance given $Z_{t-1} = z, z < 0$ is

$$\mathrm{Var}(Z_t \mid Z_{t-1} = z) = E(Z_t^2 \mid Z_{t-1} = z) - [E(Z_t \mid Z_{t-1} = z)]^2$$

$$= \frac{\Sigma_{x=0}^{\infty} (\lambda_2, \theta)_{(x-z)} (\lambda_1, \theta)_x \{p[(x-z)(x(1-p)-z) + x^2 - q(W_{(x)} + W_{(x-z)})]\} \mathrm{e}^{-2x\theta}}{\Sigma_{y=0}^{\infty} (\lambda_2, \theta)_y (\lambda_1, \theta)_{y+z} \mathrm{e}^{-2(y+z)\theta}}$$

$$-(pZ_{t-1} + \mu_v)^2.$$

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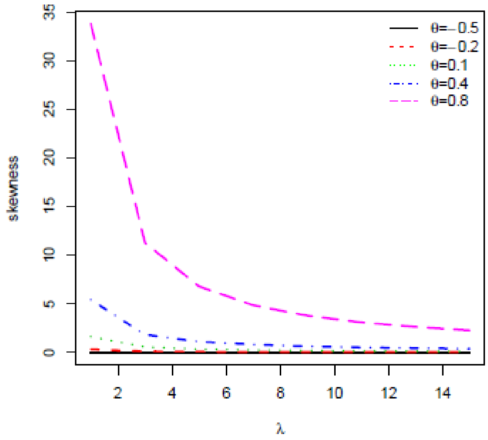
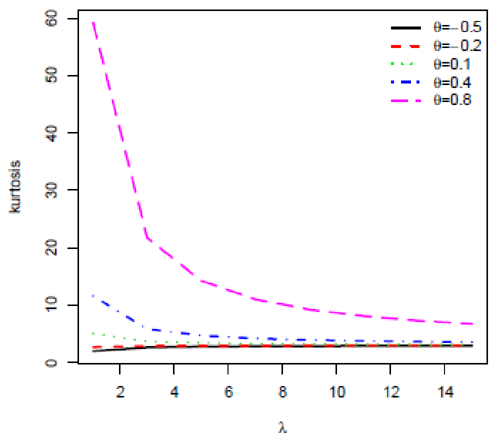
Figure 1: Skewness and kurtosis of GP distribution.

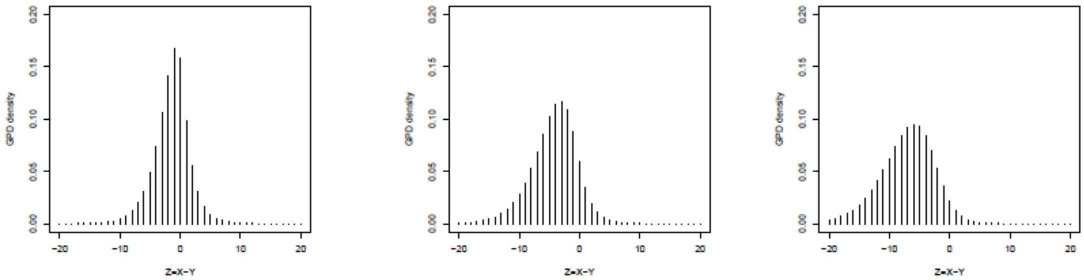


(a) Skewness of GP distribution



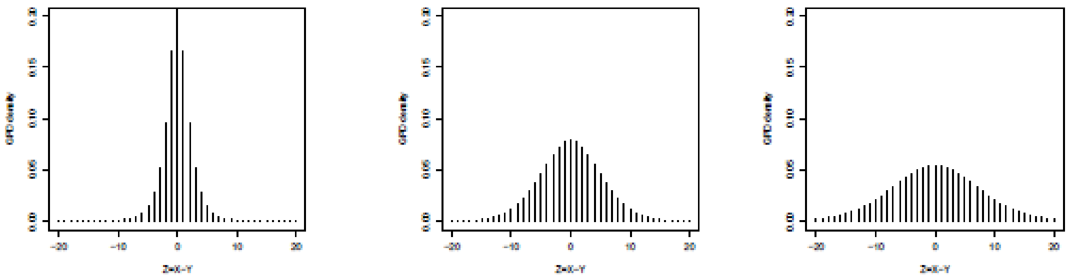
(b) Kurtosis of GP distribution

Figure 2: GPD density $\theta = 0.3, \lambda_1 = 1$ and $\lambda_2 = 2,4,6$.



(a) $\lambda_1 = 1, \theta = 0.3, \lambda_2 = 2$ (b) $\lambda_1 = 1, \theta = 0.3, \lambda_2 = 4$ (c) $\lambda_1 = 1, \theta = 0.3, \lambda_2 = 6$

Figure 3: GPD density $\theta = 0.3$ and $\lambda_1 = \lambda_2 = 1, 5, 10$.

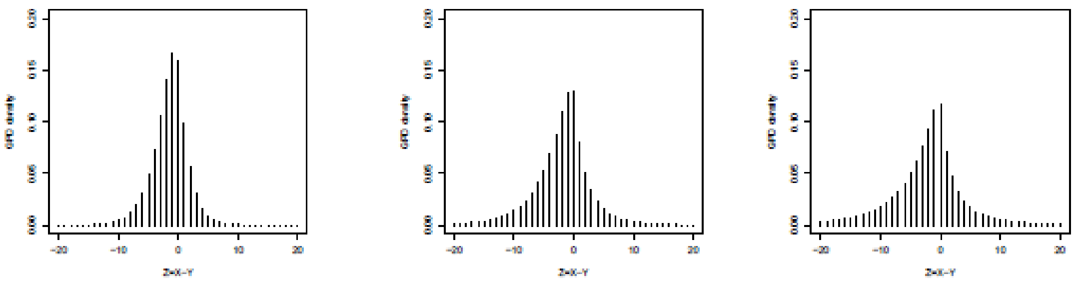


(a) $\lambda_1 = \lambda_2 = 1, \theta = 0.3$

(b) $\lambda_1 = \lambda_2 = 5, \theta = 0.3,$

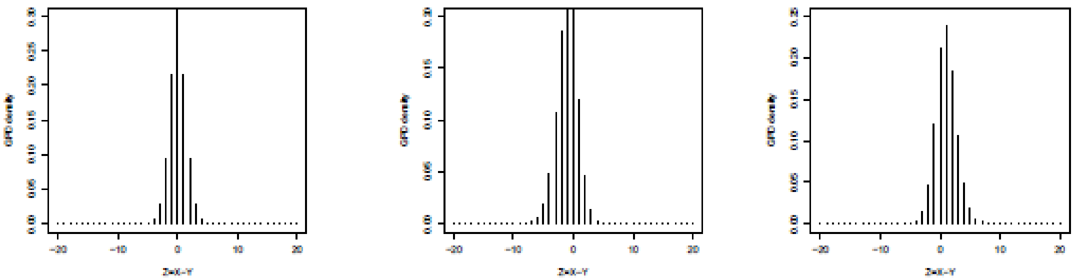
(c) $\lambda_1 = \lambda_2 = 10, \theta = 0.3$

Figure 4: GPD density $\lambda_1 = 1, \lambda_2 = 2$ and $\theta = 0.3, 0.5, 0.6$.



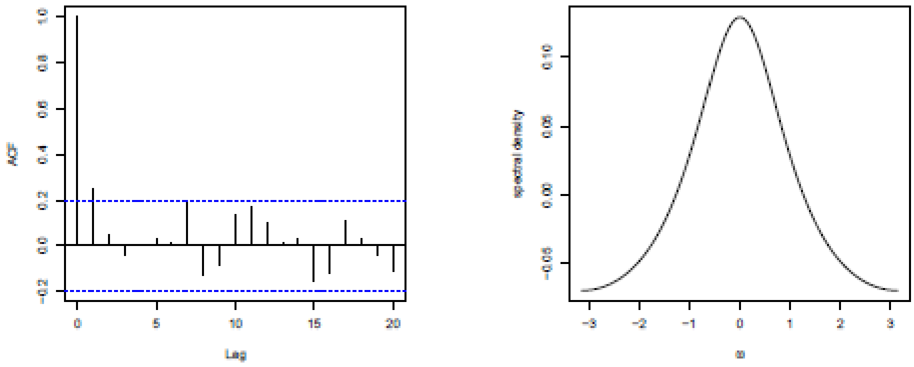
(a) $\lambda_1 = 1, \lambda_2 = 2, \theta = 0.3$ (b) $\lambda_1 = 1, \lambda_2 = 2, \theta = 0.5$ (c) $\lambda_1 = 1, \lambda_2 = 2, \theta = 0.6$

Figure 5: GPD density $\theta = 0$, $(\lambda_1, \lambda_2) = (1,1)$, $(\lambda_1, \lambda_2) = (1,2)$ and $(\lambda_1, \lambda_2) = (2,1)$.



(a) $\lambda_1 = 1, \lambda_2 = 1, \theta = 0$ (b) $\lambda_1 = 1, \lambda_2 = 2, \theta = 0$ (c) $\lambda_1 = 2, \lambda_2 = 1, \theta = 0$

Figure 6: Sample autocorrelations based on 100 simulated values of the GPDAR(1) and spectral density function for $p = 0.3, \lambda_1 = 1, \lambda_2 = 2$ and $\theta = 0.8$.



(a) Sample autocorrelation

(b) Spectral density

Figure 7: Time series plot, autocorrelation and partial autocorrelation functions for the GLOBAL Land-Ocean temperature index.

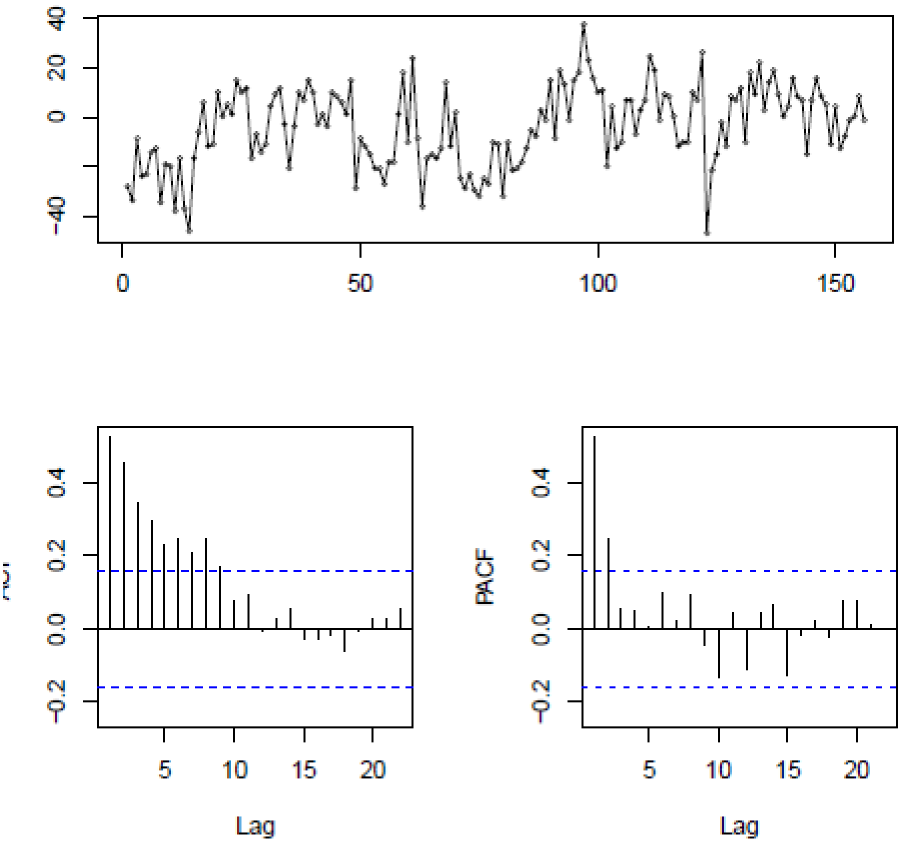


Figure 8: Sample autocorrelations of the residuals obtained from GPDAR(1) model.

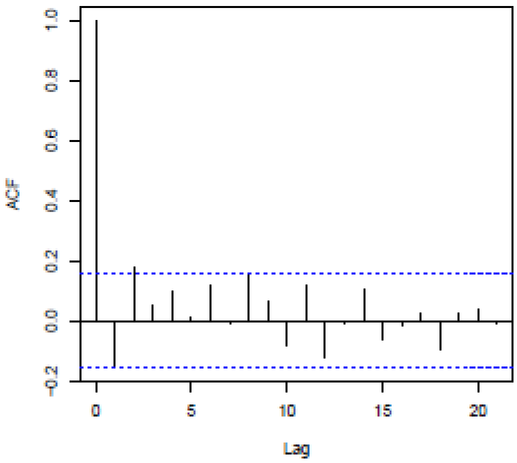


Table 1: Bias and mean squared error (in parentheses) of the moment estimates of the parameters for $p = 0.5, 0.6$ $\lambda_1 = 1, 2$, $\lambda_2 = 1, 2$, $\theta = 0.6$ and $T = 50, 100, 200$.

T	$\hat{p}$	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$\hat{\theta}$
	$p = 0.6, \lambda_1 = 1.0, \lambda_2 = 2.0$ and $\theta = 0.6$			
50	-0.0960 (0.0362)	0.1257 (2.6563)	0.3166 (4.2966)	-0.0211 (0.0188)
100	-0.0543 (0.0195)	0.0946 (2.2423)	0.2017 (3.0628)	-0.0032 (0.0130)
200	-0.0329 (0.0099)	0.1626 (2.9754)	0.2134 (3.6921)	0.0024 (0.0109)
	$p = 0.5, \lambda_1 = 1.0, \lambda_2 = 2.0$ and $\theta = 0.6$			
50	-0.0842 (0.0365)	0.1204 (2.6663)	0.2988 (4.1294)	-0.0138 (0.0168)
100	-0.0504 (0.0198)	0.1396 (2.6824)	0.2313 (3.5838)	-0.0040 (0.0125)
200	-0.0295 (0.0109)	0.1599 (2.7864)	0.1997 (3.3934)	0.0033 (0.0101)
	$p = 0.6, \lambda_1 = 2.0, \lambda_2 = 1.0$ and $\theta = 0.6$			
50	-0.0971 (0.0371)	0.3248 (5.2681)	0.1466 (3.5077)	-0.0223 (0.0192)
100	-0.0579 (0.0197)	0.1889 (2.9672)	0.0841 (2.0812)	-0.0038 (0.0140)
200	-0.0328 (0.0104)	0.2081 (3.5633)	0.1566 (2.8898)	0.0038 (0.0106)
	$p = 0.5, \lambda_1 = 2.0, \lambda_2 = 1.0$ and $\theta = 0.6$			
50	-0.0908 (0.0381)	0.2167 (2.8068)	0.0594 (1.6708)	-0.0142 (0.0168)
100	-0.0500 (0.0199)	0.2812 (3.9913)	0.1786 (3.0712)	-0.0053 (0.0133)
200	-0.0287 (0.0104)	0.1999 (3.3034)	0.1577 (2.7065)	0.0041 (0.0096)

Table 2: Descriptive statistics.

Minimum	Median	Mean	Variance	Skewness	Kurtosis	$\widehat{\rho}(1)$	Maximum
-47	-2.50	-4.07	260.8	-0.245	2.60	0.523	38

Table 3: Estimates of the parameters, MdAE and estimated quantities for the GPDAR(1) and TINAR(1) processes.

Model	Estimates	MdAE	σ_Z^2 / μ_Z	γ_1	γ_2
GPDAR(1)	$\hat{p} = 0.5233$	8.6871	-64.071	-0.2433	3.7966
	$\hat{\lambda}_1 = 3.3152$				
	$\hat{\lambda}_2 = 4.5841$				
	$\hat{\theta} = 0.6882$				
TINAR(1)	$\hat{p} = 0.5233$	8.6909	-64.071	-0.0010	0.0038
	$\hat{\lambda}_1 = 61.1691$				
	$\hat{\lambda}_2 = 63.1088$				
Empirical			-64.071	-0.2433	2.5993